

# Real-Time Public Sentiment Dashboard

Tracking public opinion about brands from Twitter/X data using Python, NLTK and Streamlit

## 1. Introduction

Brands need to know how customers feel about them as it happens, not weeks later. Social media, especially Twitter/X, is where complaints and praise surface first. This project builds a small end-to-end system that receives tweets continuously, cleans and classifies them, and shows the results on a live dashboard. It also keeps a daily sentiment log for trend tracking. The public *Twitter US Airline Sentiment* dataset (14,640 tweets about six airlines, labelled positive, neutral or negative) was used.

## 2. Abstract

Because the free X API no longer supports streaming, the labelled tweets are replayed as a live stream (a Tweepy-based real streamer with the same output format is also included). Each tweet is cleaned with NLTK, classified by a TF-IDF + Logistic Regression model, and additionally scored with NLTK VADER. Results are written to a shared file that a Streamlit dashboard reads and refreshes every few seconds, while a batch job produces daily per-brand logs. On held-out data the trained model reached **77% accuracy**, versus **54%** for VADER alone, showing that domain-specific training matters for airline complaints, which are full of sarcasm and domain vocabulary. Negative sentiment dominates overall (63% of tweets).

## 3. Tools Used

| Purpose                  | Tool                                                                               |
|--------------------------|------------------------------------------------------------------------------------|
| Language / data handling | Python 3, pandas, NumPy                                                            |
| Streaming                | Simulated replay (stream_simulator.py); Tweepy StreamingClient for real X data     |
| NLP                      | NLTK: stopwords, WordNet lemmatizer, VADER sentiment analyser                      |
| Machine learning         | scikit-learn: TF-IDF (1-2 grams), Logistic Regression, joblib for saving the model |
| Dashboard and logs       | Streamlit (auto-refreshing fragment), CSV daily logs; matplotlib for charts        |

## 4. Steps Involved in Building the Project

- **Data collection:** loaded the airline tweets dataset and inspected class balance (63% negative, 21% neutral, 16% positive).
- **Cleaning (NLTK):** lower-casing; removing URLs, @mentions and #; keeping letters only; removing stopwords *except* negations (not, no, never) so that "not happy" keeps its meaning; lemmatising words.
- **Baseline scoring:** ran VADER on the raw text (it uses capitals, "!" and emoji) and compared it with human labels: 54% accuracy, weak on sarcasm and complaints.
- **Model training:** TF-IDF features (up to 10,000, unigrams and bigrams) with class-balanced Logistic Regression on an 80/20 stratified split: 77% accuracy, negative-class F1 0.85. The model is saved and reused by the stream.
- **Streaming pipeline:** every incoming tweet is timestamped, cleaned, classified and VADER-scored, then appended to data/stream/live\_tweets.csv.
- **Live dashboard:** Streamlit shows tweet count, sentiment shares, a net-sentiment KPI ((positive - negative) / total), a time trend, a per-brand comparison and the latest tweets, refreshing automatically.
- **Daily logs:** a batch script aggregates the stream into logs/daily\_sentiment.csv (per day and brand: counts, average VADER score, net sentiment), which can also be loaded into Tableau.

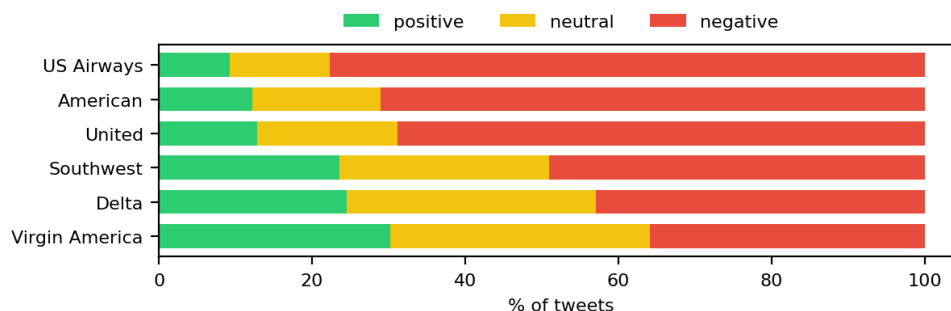


Figure 1. Sentiment share by airline in the dataset (sorted by negative share).

## 5. Results

| Method                       | Accuracy | Note                                          |
|------------------------------|----------|-----------------------------------------------|
| NLTK VADER (no training)     | 54%      | Over-predicts positive on complaints          |
| TF-IDF + Logistic Regression | 77%      | Held-out 20%; neutral class weakest (F1 0.61) |

## 6. Conclusion

The project delivers the three requested outputs: a live sentiment dashboard, a Python streaming and NLP pipeline, and daily sentiment logs. Its main lesson is that a general-purpose lexicon (VADER) is not enough for a specific domain, while a simple trained model improved accuracy by 23 points. **Limitations:** the data is from 2015 and covers airlines only; the replayed stream overlaps the model's training tweets, so the 77% held-out score is the honest figure; sarcasm and the neutral class remain difficult. **Future work:** a transformer model such as DistilBERT, real X API ingestion, message queues (Kafka) for scale, and drift monitoring.